

[56/57]

BCA SEMESTER-II (Old Batch)

External Examination 2022

US02SBCA21-Mathematics
/US02SBCA51

Date: -24-08-2022

Time: - 02:00pm to 04:00pm

Marks: -70

Q.1 Multiple Choice Questions

[10]

1. Edges connecting the same end points are called
(a) trivial graph (b) multigraph (c) loops (d) multiple edges
2. A graph with one vertex and no edges is called
(a) trivial graph (b) multigraph (c) null graph (d) multiple edges
3. A graph with no vertices and no edges is called ...
(a) trivial graph (b) multigraph (c) null graph (d) multiple edges
4. A graph that can be drawn in a plane or on a sphere so that its edge do not cross is said to be
(a) Complete graph (b) simple graph (c) planar graph (d) Non-planar graph
5. A map is a particular representation of a finite planar multigraph.
(a) planar (b) non planar (c) complete (d) simple
6. For connected map, the relation between number of vertices V, number of edges E and number of regions R is
(a) $R + E - V = 2$ (b) $V - E - R = 2$ (c) $E - R + V = 2$ (d) $V - E + R = 2$
7. If $u = (1,2)$ and $v = (3,2)$ then $u + v =$
(a) (4,3) (b) (4,4) (c) (3,4) (d) (1,4)
8. If $u = (a, b)$ and $v = (c, d)$ then $u \bullet v =$
(a) $(ab + cd)$ (b) $(ac+bd)$ (c) $ad + bc$ (d) none
9. is the value dividing the data into two equal parts.
(a) mode (b) median (c) mean (d) none
10. The observation whose frequency is highest is called
(a) mean (b) median (c) mode (d) none

Q.2 State True /False and Fill in the Blanks

[08]

1. Bipartite graph is denoted by
2. A graph G is if each vertex is connected to every other vertex.
3. Complete graph is denoted by
4. A spanning tree T of graph contains all the of G.
5. The norm of a vector d is denoted by $\|w\|$ State True or False.
6. If $u=(1,2)$ and $v=(2,2)$ then $u + v = (3,3)$ State True or False
7. The observation whose frequency is highest is called mode. State True or False
8. Mode is not the measure of Dispersion. State true or false.

(1)

(P.T.O.)

Q.3 Write a short answer for given questions (10 out of 12)

[20]

1. Draw a picture of given values

(a) $G_1 = (V_1, E_1)$, where $V_1 = \{a, b, c, d, e\}$ and $E_1 = \{ab, bc, ac, ad, de\}$.

(b) $G_2 = (V_2, E_2)$, where $V_2 = \{P, Q, R, S, T\}$ and $E_2 = \{PQ, PR, PS, PT, TR, PR\}$.

2. Define Regular Graph and Complete Graph.

3. Define Null Graph and trail graph.

4. Define Mean with example.

5. Write a formula for Median.

6. Write a Euler's formula and give one example.

7. Find x and y :

(i) if $(x, 7) = (4, x + y)$ (ii) if $(20, y) = x(4, 3)$

8. If $u = (1, 2, 3, -4)$, $v = (5, -6, 7, 8)$, $w = (1, -2, 5, 3)$ and $k=3$ then find :

(i) $(u + v) \cdot w$ (ii) $u \cdot v + v \cdot w$

9. Determine k such that $\|u\| = \sqrt{39}$ where $u = (1, k, -3, -5, 1)$

10. The intelligence quotients (IQ's) of 10 boys is given below:

70, 120, 110, 101, 88, 83, 95, 98, 107, 100 Find the Mean

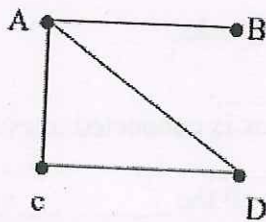
11. Define Planar Graph with example.

12. Write a formula for Bipartite Graph.

Q.4 Long Questions Answers (4 out of 8)

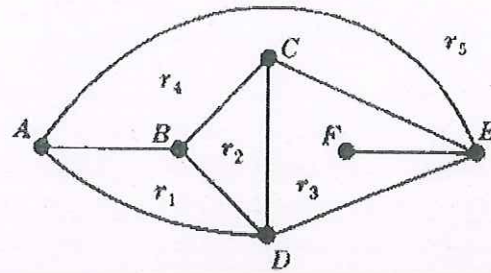
[32]

1. Draw cut point graphs and Bridges graph from given graph.

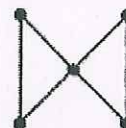
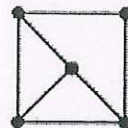


2. Draw the complete graph. (1) K_4 (2) K_5 (3) K_3 (4) K_2

3. Find the degree of region, closed path and cycle from given graph.



4. Find the number of spanning tree from given graph.



5. Find the inverse of $A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{pmatrix}$

6. Find $AB+BA$ where $A = \begin{pmatrix} 2 & 6 & 2 \\ 5 & -1 & 3 \\ 3 & 1 & -2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 4 & 5 \\ 0 & -1 & 7 \\ 4 & -3 & 2 \end{pmatrix}$

7. Calculate the mean, median, mode for the following data.

xi	fi
10	14
30	23
50	27
70	21
90	15

8. Calculate the mean, median and mode for given data.

Marks :	0- 10	10-20	20-30	30-40	40-50	50-60
No. of students:	12	18	27	20	17	6

